

# All-prime tensor boundary weights and the original support receiver

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## 1 The object and the part of weight control computed here

The absolute base remains the whole spectrum of the original globalization semiring. For a bounded distributive lattice  $L$  and a ring  $R$ , its coefficient-and-support object is

$$G_L(R) = \{(0, \lambda) : \lambda \in L\} \cup \{(a, 1_L) : a \in R\}, \quad \tau = (0, 0_L), \quad e = (0, 1_L). \quad (\text{BW1})$$

Addition uses coefficient addition and lattice join; multiplication uses coefficient multiplication and lattice meet. The overlap in this union is exactly  $e$ . In particular, the base for  $L = \{0, 1\}$  and  $R = \mathbb{Z}$  retains  $(\tau) \subsetneq (e) \subsetneq (p)$ . We do not replace this space by its open Boolean localization or by the closed arithmetic stratum.

The full function-valued support faces of the programme have an actual scalar boundary map: on each place, evaluate at supported zero and integrate with the original self-dual additive Haar measure. Its precise definition and original rational action are AB1–5, AB20–24 of [1]. This paper computes weights on all tensor powers of that specific boundary receiver and their rational group homology. It also gives the exact connecting map back into the original test-space kernel. It does not identify this boundary homology with the interior cohomology of the zeta programme.

The reason to compute tensors before quotients is mathematical: Deligne’s weight argument uses tensor constituents and coinvariants in its cohomological trace calculation, followed by duality [2, 1.3.13, 1.4.1–1.5.3, 1.8.4, 3.3.1–3.3.6]. Below, mixed tensor coinvariants survive which are killed by the tensor power of the two-endpoint quotient. All their actual weights and multiplicities are calculated. No Frobenius on this coefficient space is assumed: the operator used here is the original idele dilation.

## 2 The full boundary and all its tensor coinvariants

Write  $\mathcal{P}$  for all rational primes and  $\Omega = \{0, 1\}^{\{\infty\} \sqcup \mathcal{P}}$ , with product topology. For a finite  $F \subset \mathcal{P}$  put

$$\Omega_F = \{0, 1\}^{\{\infty\} \sqcup F}, \quad B_F = \mathbb{C}^{\Omega_F}, \quad j_{FF'} e_A = \sum_{C \subseteq F' \setminus F} e_{A \cup C}. \quad (\text{BW2})$$

Thus  $j_{FF'}$  is exactly tensoring by  $(1, 1)$  at every new prime. Here  $e_A$  is a coordinate vector, not the supported scalar zero  $e$ . It is the characteristic function of the cylinder with that coordinate pattern. A locally constant function on a compact space has finite image. On this product, each such function depends on finitely many coordinates: cover the space by cylinders on which it is constant, take a finite subcover, and take the union of their coordinate sets. Consequently

$$B = \varinjlim_F B_F = \text{LC}(\Omega, \mathbb{C}), \quad V_r = B^{\otimes \mathbb{C}^r} = \text{LC}(\Omega^r, \mathbb{C}) \quad (r \geq 1). \quad (\text{BW3})$$

The tensor is algebraic. The second equality follows at finite stages by the characteristic-function bases and then by the same finite cylinder argument. The diagonal family of stages  $B_F^{\otimes r}$  is cofinal among all choices of  $r$  finite stages.

For  $\omega = (\varepsilon_{v,j}) \in \Omega^r$  set  $m_v(\omega) = \sum_{j=1}^r \varepsilon_{v,j}$ . The actual diagonal action of  $\gamma = \pm \prod_p p^{n_p} \in \mathbb{Q}^\times$  is

$$(D_\gamma^{(r)} f)(\omega) = \prod_p p^{n_p(m_\infty(\omega) - m_p(\omega))} f(\omega). \quad (\text{BW4})$$

Only finitely many  $n_p$  are nonzero. This is obtained by tensoring AB5; no factor in the original action has been removed.

Define the following closed set and its finite stages:

$$Z_r = \{\omega : m_p(\omega) = m_\infty(\omega) \text{ for all } p\}, \quad Z_{r,F} = \{\omega \in \Omega_F^r : m_p = m_\infty \ (p \in F)\}. \quad (\text{BW5})$$

**Theorem 1** (Every all-prime tensor coinvariant). *Restriction gives a surjection  $\rho_r : V_r \rightarrow \text{LC}(Z_r, \mathbb{C})$  with*

$$\ker \rho_r = \sum_{p \in \mathcal{P}} (D_p^{(r)} - I) V_r. \quad \text{Hence} \quad (V_r)_{\mathbb{Q}^\times} = \text{LC}(Z_r, \mathbb{C}). \quad (\text{BW6})$$

The sum consists of finite sums. Also  $(V_r)^{\mathbb{Q}^\times} = 0$ . Writing  $\binom{[r]}{m}$  for the set of  $m$ -element subsets of  $\{1, \dots, r\}$ , there is the exact decomposition

$$Z_r = \prod_{m=0}^r Z_{r,m}, \quad Z_{r,m} = \binom{[r]}{m} \times \prod_{p \in \mathcal{P}} \binom{[r]}{m}, \quad |Z_{r,F}| = \sum_{m=0}^r \binom{r}{m}^{|F|+1}. \quad (\text{BW7})$$

Each component is both open and closed in  $Z_r$ .

*Proof.* Every multiplier  $D_p^{(r)} - I$  vanishes on  $Z_r$ . Conversely let  $f \in V_r$  vanish there. Its nonzero locus is a compact open and closed set, disjoint from  $Z_r$ . Every point of that locus has  $m_p \neq m_\infty$  for at least one prime. Finitely many such primes cover the locus. Partition it into finitely many disjoint cylinders subordinate to this cover. On one cylinder  $U$  choose its indicated prime  $p$ . The locally constant multiplier  $u_p = p^{m_\infty - m_p} - 1$  is nonzero there, so  $g = \mathbf{1}_U f / u_p$ , extended by zero, lies in  $V_r$  and  $(D_p^{(r)} - I)g = \mathbf{1}_U f$ . Summing proves the kernel identity.

Restriction is onto. A locally constant function on the compact closed subset  $Z_r$  has a finite partition into relatively open and closed sets of constant values. Each piece is the intersection with an open and closed set of  $\Omega^r$ : choose cylinders inside its relative open neighbourhood and use a finite cover of that compact piece. Successive differences make the chosen extensions disjoint without altering their intersections with  $Z_r$ . Extending the values there and assigning zero elsewhere gives the desired preimage.

Relations from primes generate the relations of the whole group:  $D_{ab} - I = D_a(D_b - I) + (D_a - I)$  and  $D_{p^{-1}} - I = -D_{p^{-1}}(D_p - I)$ ; the sign acts trivially. This proves the coinvariant assertion.

An invariant  $f$  must be supported on  $Z_r$ . The latter has empty interior: after fixing finitely many coordinates, choose a new prime and a column with sum different from the prescribed infinity sum. Since  $r \geq 1$ , such a column exists. Every nonzero locally constant function is nonzero on some open cylinder, so no such  $f$  exists. Finally, for fixed  $m$ , each column records exactly an  $m$ -element subset. The choices at all places are independent. This proves BW7, including the finite count.  $\square$

**Proposition 2** (Exact loss under the endpoint tensor quotient). *Let  $\epsilon : B \rightarrow \mathbb{C}^2$  evaluate at the two constant patterns  $0^\infty, 1^\infty$ . Its tensor power factors as*

$$V_r \xrightarrow{\rho_r} \text{LC}(Z_r) \xrightarrow{\text{res}_{E_r}} \mathbb{C}^{E_r} = (\mathbb{C}^2)^{\otimes r}, \quad E_r = \{(\omega_1, \dots, \omega_r) : \omega_j \in \{0^\infty, 1^\infty\}\}. \quad (\text{BW8})$$

For  $r \geq 2$  the kernel of the second map is infinite dimensional. For  $r = 2$  and any prime  $p$ , the actual finite-stage tensor

$$e_{\{\infty\}} \otimes e_{\{p\}} \in B_{\{p\}}^{\otimes 2} \quad (\text{BW9})$$

has a nonzero image in  $\text{LC}(Z_2)$  and zero image under  $\epsilon^{\otimes 2}$ .

*Proof.* Each endpoint tuple is balanced, so BW8 is the stated restriction. For BW9 the infinity column is  $\{1\}$  and the  $p$  column is  $\{2\}$ ; their sums are both one. Complete all other prime columns arbitrarily with sum one. This gives a point of  $Z_2$  where BW9 has value one. On an endpoint tuple every prime column equals the infinity column, so BW9 vanishes there. For general  $r \geq 2$  choose  $0 < m < r$  and a prime column different from the infinity column, with both of size  $m$ . This defines a nonempty cylinder in  $Z_{r,m}$  disjoint from  $E_r$ . Refining it at arbitrarily many other primes yields arbitrarily many linearly independent disjoint cylinder functions in the kernel. The whole locally constant space has countable dimension, because its finite stages form a countable exhaustion. Thus this kernel has countably infinite dimension.  $\square$

### 3 Compatible prime-weight filtrations before the quotient

The finite prime weights of the actual operator on  $V_r$  are

$$w_p(\omega) = 2(m_\infty(\omega) - m_p(\omega)), \quad D_p^{(r)} f = p^{w/2} f \quad \text{on } V_{r,p,w} = \mathbf{1}_{\{w_p=w\}} V_r. \quad (\text{BW10})$$

Every even  $w$  between  $-2r$  and  $2r$  occurs. Therefore the following is a filtration of the original module, defined at all primes:

$$W_n^p V_r = \bigoplus_{w \leq n} V_{r,p,w}, \quad \Pi_{p,2a} = \prod_{\substack{-r \leq b \leq r \\ b \neq a}} \frac{D_p^{(r)} - p^b I}{p^a - p^b} \quad (-r \leq a \leq r). \quad (\text{BW11})$$

The denominators are nonzero since  $p > 1$ . On each cylinder the polynomial is one for its named exponent and zero for every other exponent, proving the formula. These projectors commute at different primes and commute with every later support inclusion after that prime has been included. Thus every finite collection of these filtrations is compatible with the actual direct limit. Tensoring coefficient directions adds  $w_p$  exactly. BW9 is the explicit cancellation of weights  $+2$  and  $-2$  into weight zero. The original operator, rather than a scalar change of it, gives these weights.

At a finite stage the projection to simultaneous weight zero is restriction to  $Z_{r,F}$  followed by extension by zero. Under a new prime this is not the old projection followed by the unrestricted support inclusion: one must retain exactly the new columns with the same sum  $m$ . The induced map on coinvariants is

$$\text{LC}(Z_{r,F}) \longrightarrow \text{LC}(Z_{r,F'}), \quad f \longmapsto f \circ \text{pr}_{F'F}. \quad (\text{BW12})$$

On the  $m$  component each coordinate is repeated exactly  $\binom{r}{m}^{|F' \setminus F|}$  times. Hence in the original counting norm its squared norm is multiplied by that precise integer. Before restriction to balanced columns, the squared-norm multiplier is  $2^{r|F' \setminus F|}$ . Both statements follow by counting the actual extended columns; no factor is absorbed in the transition map.

### 4 All-degree rational homology and actual idele weights

Let  $E = \bigoplus_{p \in \mathcal{P}} \mathbb{C} \theta_p$ , with primes ordered by their ordinary increasing order. Use algebraic group homology.

**Theorem 3** (Complete homology of the boundary tensor). *For every  $r \geq 1$  and  $j \geq 0$  there is the natural isomorphism*

$$H_j(\mathbb{Q}^\times, V_r) = \text{LC}(Z_r, \mathbb{C}) \otimes_{\mathbb{C}} \Lambda^j E. \quad (\text{BW13})$$

Let  $a = (a_v)$  be an idele, with original modulus  $|a| = |a_\infty|_\infty \prod_p |a_p|_p$ . On the summand  $\text{LC}(Z_{r,m}) \otimes \Lambda^j E$ , its original unweighted dilation acts exactly by

$$D_a^{(r)} = |a|^m I. \quad (\text{BW14})$$

Thus the global modulus weights, in the convention  $|a|^{w/2}$ , are exactly  $w = 0, 2, \dots, 2r$ . For  $j = 0$  the two extreme components have multiplicity one; every interior component has countably infinite dimension. For each  $j \geq 1$ , every component has countably infinite dimension.

*Proof.* For a finite prime set  $F$ , put  $R_F = \mathbb{C}[t_p^{\pm 1} : p \in F]$ . The sequence  $(t_p - 1)_{p \in F}$  is regular: successive quotients are Laurent polynomial rings on the remaining variables, and multiplication by the next  $t_p - 1$  is injective. Its Koszul resolution is consequently free and exact; this also follows inductively by the mapping cone of the one-variable resolution. The Laurent ring  $R$  on all primes is free over each  $R_F$ , with basis the Laurent monomials in the other variables. Extending the finite resolution to  $R$  therefore resolves  $R/(t_p - 1 : p \in F)$ . The filtered union of these augmented complexes resolves  $R/(t_p - 1 : p \in \mathcal{P}) = \mathbb{C}$ . Filtered unions preserve exactness: an element and its proposed preimage or relation are represented at a finite stage.

After tensoring with  $V_r$  the resulting chain complex is the union of

$$\begin{aligned} C_j(F) &= B_F^{\otimes r} \otimes \Lambda^j \mathbb{C}^F, \\ d(b \otimes \theta_{p_1} \wedge \cdots \wedge \theta_{p_j}) &= \sum_{h=1}^j (-1)^{h-1} (D_{p_h}^{(r)} - I)b \\ &\quad \otimes \theta_{p_1} \wedge \cdots \widehat{\theta_{p_h}} \cdots \wedge \theta_{p_j}. \end{aligned} \quad (\text{BW15})$$

Every coefficient vector and every exterior vector occurs in some common finite stage. The support inclusions commute with the old prime actions, so BW15 is a chain inclusion into every larger stage.

Decompose at stage  $F$  into the original coordinate lines of  $\Omega_F^r$ . On a line set  $c_p = p^{m_\infty - m_p} - 1$ . If one  $c_p \neq 0$ , the degree-raising map  $c_p^{-1} \theta_p \wedge (-)$  satisfies  $dh + hd = I$ , by the exterior contraction identity. On the balanced lines every  $c_p = 0$ , so the differential is zero. Projection to those lines therefore gives exactly

$$H_j(C_\bullet(F)) = \text{LC}(Z_{r,F}) \otimes \Lambda^j \mathbb{C}^F. \quad (\text{BW16})$$

For a larger stage, projecting an included balanced cycle discards the unbalanced new columns and leaves precisely BW12, together with the ordinary exterior inclusion. Homology commutes with this filtered union: a cycle and a proposed bounding chain each have a finite stage representative. Hence BW16 gives BW13 for the positive rational group. The sign subgroup has order two, acts trivially, and its trivial module over  $\mathbb{C}$  is projective by the averaging idempotent  $(1 + [-1])/2$ . Tensoring the resolution with this projective resolution concentrated in degree zero proves BW13 for all of  $\mathbb{Q}^\times$ .

The actual local integral scales by  $|a_v|_v$  and the local evaluation is unchanged. On  $V_r$  the multiplier is therefore  $\prod_v |a_v|_v^{m_v}$ . The product is finite away from infinity, because an idele has  $|a_p|_p = 1$  at all but finitely many primes. This action commutes with rational dilation, acts trivially on the named group generators in BW15, and on  $Z_{r,m}$  is exactly  $\prod_v |a_v|_v^m = |a|^m$ . This proves BW14 without a new operator. BW7 gives the multiplicities in degree zero. For  $j \geq 1$ ,  $\Lambda^j E$  has a countably infinite basis of  $j$ -element prime sets. This proves all the remaining multiplicities.  $\square$

This is weight control on the computed boundary homology. It is not the assertion that cohomological degree itself adds a weight: the factor in BW14 has been calculated and depends on  $m$ , not on  $j$ . Nor is  $H_j(\mathbb{Q}^\times, V_r)$  being relabelled  $H^j(\mathbb{Q}^\times, V_r)$ . The latter has a different degree-zero group, already visible in BW6 and its invariant statement.

## 5 The connecting map from these weights to the original kernel

Let

$$\mathcal{T} = \mathcal{S}(\mathbb{R}) \otimes \bigotimes_p' (\mathcal{S}(\mathbb{Q}_p), \mathbf{1}_{\mathbb{Z}_p}), \quad b = \bigotimes_v (f \mapsto (f(0), \int f)) : \mathcal{T} \rightarrow B. \quad (\text{BW17})$$

This is the original algebraic restricted Schwartz tensor product. At each place its two displayed functionals are independent. At a finite prime,  $\mathbf{1}_{\mathbb{Z}_p}$  and  $\mathbf{1}_{p\mathbb{Z}_p}$  give moment columns  $(1, 1)$  and  $(1, p^{-1})$ , whose determinant is nonzero. At infinity, a nonzero nonnegative smooth compact function supported away from zero has zero evaluation and positive integral; combine it with a smooth compact function of nonzero value at zero. These give an explicit two-dimensional section at every place. Tensoring finite sections proves that  $b$  is onto. Its compatibility with the restricted-product reference functions follows from their moments  $(1, 1)$ , including every Haar factor.

Put  $\mathcal{T}_r = \mathcal{T}^{\otimes r}$  and

$$J_r = \ker(b^{\otimes r}) = \sum_{i=1}^r \mathcal{T}^{\otimes(i-1)} \otimes \ker b \otimes \mathcal{T}^{\otimes(r-i)}. \quad (\text{BW18})$$

The equality follows by choosing a vector-space section of the surjective linear map  $b$  and expanding the tensor of its direct sum with its kernel. Both sides are independent of that choice. The maps are equivariant for rational and idele dilation, so

$$0 \longrightarrow J_r \longrightarrow \mathcal{T}_r \xrightarrow{b^{\otimes r}} V_r \longrightarrow 0 \quad (\text{BW19})$$

is an exact sequence of the original modules. Its homology sequence contains the exact connecting maps

$$H_j(\mathbb{Q}^\times, \mathcal{T}_r) \longrightarrow \bigoplus_{m=0}^r \text{LC}(Z_{r,m}) \otimes \Lambda^j E \xrightarrow{\partial_r} H_{j-1}(\mathbb{Q}^\times, J_r). \quad (\text{BW20})$$

For this connecting map the sign subgroup must be kept as well: it acts trivially on  $V_r$  but can act nontrivially on  $\mathcal{T}_r$  and  $J_r$ . Write  $\iota = U_{-1}^{\otimes r}$  and retain the two exact projectors  $P_+ = (I + \iota)/2$ ,  $P_- = (I - \iota)/2$ . They sum to the identity, multiply to zero and commute with every rational and idele action. Taking  $P_+$  is exact on equivariant vector-space sequences, since an arbitrary lift can be replaced by its  $P_+$  image. The odd summand has zero homology for the full group: the trivial  $\mathbb{C}[\{\pm 1\}]$  module is its even projective summand, whose tensor with an odd module is zero. Tensoring with the positive-prime resolution proves this in every degree for  $\mathbb{Q}^\times$ . Thus the original full-group homology is calculated by the positive-prime Koszul complex on the even summands, with this explicit comparison from the entire coefficient module.

To calculate  $\partial_r$ , represent a cycle in BW15 and lift its finitely many coefficient vectors to  $\mathcal{T}_r$ . Replace each lift  $t$  by the exact even lift  $P_+ t = (t + U_{-1}^{\otimes r} t)/2$ , whose boundary image is unchanged. Apply the same differential BW15 to these lifts. Its boundary image is zero

because the original vector was a cycle, so the result belongs to  $J_r$  and represents  $\partial_r$ . Changing a lift adds a chain in  $J_r$  and changes the answer by its differential. Changing a cycle by a boundary changes the answer by a boundary by  $d^2 = 0$ . These observations prove the formula, its well-definedness and the usual exactness directly. Every operation commutes with idele dilation. Therefore

$$D_a \partial_r(c) = |a|^m \partial_r(c) \quad \text{for } c \in \text{LC}(Z_{r,m}) \otimes \Lambda^j E. \quad (\text{BW21})$$

This computes the weight of every nonzero class actually reached by this connecting map; it does not assert that all its classes are nonzero or that it exhausts the interior.

In degree one, a vector  $b_0$  fixed by  $D_p^{(r)}$  gives the cycle  $b_0 \otimes \theta_p$ . For any original test lift  $t$ ,

$$\partial_r[b_0 \otimes \theta_p] = [(U_p^{\otimes r} - I)t] \quad \text{in } (J_r)_{\mathbb{Q}^\times}, \quad b^{\otimes r} t = b_0. \quad (\text{BW22})$$

For BW9 this applies at its named prime  $p$ : its two factors have eigenvalues  $p$  and  $p^{-1}$  and their product is one. It supplies the exact map into the original kernel for the mixed class erased by the endpoint tensor quotient.

Formula BW22 allows any lift because  $t - P_+ t$  belongs to  $J_r$ , and  $(U_p^{\otimes r} - I)(t - P_+ t)$  is already a coinvariant relation inside that original kernel. The factor  $1/2$  and the odd summand have thus both been accounted for explicitly.

## 6 Keeping the support lattice on every receiving map

For a complex vector space  $M$  use the labelled coefficient semimodule

$$G_L(M) = \{(0, \lambda) : \lambda \in L\} \cup \{(v, 1_L) : v \in M\}. \quad (\text{BW23})$$

Its addition and the action of  $G_L(\mathbb{C})$  use precisely coefficient addition/scalar multiplication and join/meet of the labels. For every linear map  $f : M \rightarrow N$ , define  $G_L(f)(0, \lambda) = (0, \lambda)$  and  $G_L(f)(v, 1_L) = (f(v), 1_L)$ . Checking the two operations shows this is a semimodule map; it preserves all labels, including  $e$  and  $\tau$ . This is a coefficient receiver over the retained geometric base, not a replacement of that base by  $\text{Spec } G_L(\mathbb{C})$ .

In particular  $G_L(\rho_r)$  is onto and its fibres are exact: all labels  $(0, \lambda)$  with  $\lambda \neq 1_L$  have singleton fibres, while the fibre at  $(h, 1_L)$  is

$$\{(f, 1_L) : f|_{Z_r} = h\}. \quad (\text{BW24})$$

For  $h = 0$  this fibre contains  $e$  and every supported amplitude vanishing on  $Z_r$ , and contains no other zero label. Thus the amplitude relation in BW6 never identifies supported zero with  $\tau$ . Likewise  $G_L(W_n^p V_r)$  is the exact labelled filtration subobject, retaining the entire zero-label lattice at every level. Apply the same explicit functor to BW20–22 and their actual linear maps. On a supported vector its weight factor is the unchanged scalar  $|a|^m$ ; on a zero label the label is fixed. Every square commutes because the original linear square does and every map is the identity on the labels.

## 7 Consequence for the continuing weight construction

The first tensor quotient has two endpoint classes. In every higher tensor order there are further mixed classes, whose complete support space, rational homology and original idele weights are BW7, BW13 and BW14. The map from them into the actual test-space kernel is BW20–22. Taking the endpoint quotient first deletes these classes, with the explicit nonzero

lost class BW9. Consequently the tensor and cohomological stages of the weight construction must use BW19 before making that quotient.

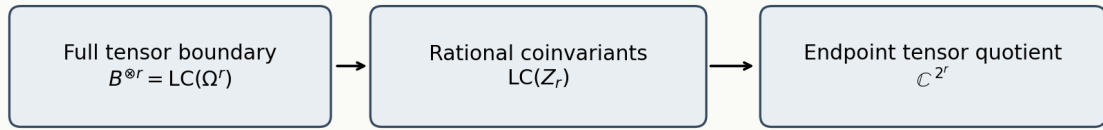
Deligne’s image-purity theorem concerns an actual map  $H_c^i \rightarrow H^i$  and follows from the upper direct-image weight theorem together with duality [2, 3.3.1, 3.3.5–3.3.6]. The present result computes the full tensor boundary term and its connecting map in our existing object. The construction of the corresponding interior image and its weight control is still the next mathematical task. No unknown purity hypothesis has been used to prove any formula above.

## References

- [1] Programme source, *Full prime-support reconstruction*, pinned complete proof source, AB1–5, AB20–24 and ME1–7. [Public complete TeX proof](#). The source bytes and the named formulas were checked for this derivation.
- [2] Pierre Deligne, *La conjecture de Weil. II*, Publications Mathématiques de l’IHÉS **52** (1980), 137–252, [original publication](#), doi:10.1007/BF02684780. Source reading uses the retained S20 French TeX transcription and separately identifies historical English translation errors; neither is represented as author-original TeX.
- [3] Alain Connes, *Trace formula in noncommutative geometry and the zeros of the Riemann zeta function*, Selecta Math. **5** (1999), 29–106, [original author source](#), Section III. The local evaluation/integration and original adelic action are the source conventions received through AB. The tensor balanced-space and homology calculations here are programme derivations, not assertions attributed to Connes or Deligne.

## Mixed tensor weights before the endpoint quotient

The actual all-prime boundary: every place retains its evaluation / integral choice.



$Z_r$ :  $m_p = m_\infty$  at every prime, where  $m_v = \sum_{j=1}^r \varepsilon_{v,j}$ .

On its  $m$  component, the original idele action is exactly  $|a|^m$ .

### Tensor order two, with one named prime $p$

The four balanced choices with one integral in each column:

|                        | p-column = {1}                                                                                           | p-column = {2}                                                                                           |                                                   |
|------------------------|----------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------|---------------------------------------------------|
| $\infty$ -column = {1} | <b>Endpoint value retained</b><br>also has mixed continuations                                           | <b>Mixed class survives in <math>\text{LC}(Z_2)</math></b><br>value is erased by $\mathbb{C}^4$ receiver | Each cell has modulus character $ a $ (weight 2). |
| $\infty$ -column = {2} | <b>Mixed class survives in <math>\text{LC}(Z_2)</math></b><br>value is erased by $\mathbb{C}^4$ receiver | <b>Endpoint value retained</b><br>also has mixed continuations                                           |                                                   |

Example:  $e_{\{\infty\}} \otimes e_{\{p\}}$  has prime eigenvalues  $p \cdot p^{-1} = 1$ .

At every other prime either one-element column is allowed; all those mixed choices remain.

$$H_j(\mathbb{Q}^\times, B^{\otimes r}) = \text{LC}(Z_r) \otimes \wedge_{\mathbb{C}}^{j_C}(\text{primes}).$$

BW6-14 prove these maps and weights. BW20-22 give the connecting map into the original test-space kernel.

Figure 1: Actual boundary tensor before the endpoint quotient. BW5-9 prove the balanced configurations and the lost class; BW13-14 compute the homology and its idele weights; BW20-24 give the connecting map and preserve the full coefficient support. The support labels are not extra points of the arithmetic prime set.